

Arbitrage Trading in the I-SEM

Predicting Short-Term Price Spreads and Evaluating Trading Strategy Performance

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1. Executive Summary

This report investigates the extent to which fundamental market variables of demand, wind, solar generation, interconnector flows and system constraints, explain and predict short-term price differences between the Integrated Single Electricity Market's (I-SEM) consecutive auction markets. The aim is to identify which market transition offers the **best risk-adjusted profitability**. Using four years of half-hourly data provided by Energia, covering over 70,000 delivery intervals from January 2022 to December 2025, we develop predictive models and translate their output into a structured arbitrage trading strategy.

Key Finding (prediction quality): DAM → IDA1 is the most predictable transition with 70.2% accuracy & AUC 0.77. However, it is not the most profitable in €/MWh terms.

Key Finding (profit): DAM → BM is the most profitable transition (€33.75/MWh, €480,887 total P&L). It was able to complete 4,258 out of 13,783 possible trades with a 70.36% hit rate.

Recommendation: If the objective is to **maximise trading profit**, prioritise **DAM → BM** with confidence filtering and drawdown monitoring. If the objective is to **maximise predictive performance**, **DAM → IDA1** is the best modelling target, but profitability depends on trade filtering and cost assumptions.

2. Market Background and Problem Definition

The I-SEM is the wholesale electricity market for Ireland and Northern Ireland, operated by SEMO. It comprises a series of sequential auction markets, each designed to give traders the opportunity to refine their positions when real-time delivery approaches and new information becomes available. The principal markets, in chronological order of gate closure, are the **Day-Ahead Market (DAM)**, three **Intraday Auctions (IDA1, IDA2, IDA3)**, and the real-time **Balancing Market (BM)**. Arbitrage in this context refers to the practice of buying or selling electricity in one market and offsetting in a later market, with the intent of profiting from the price difference (the spread) between them. The profitability of such a strategy depends entirely on the ability to predict the direction and magnitude of these spreads before they are observed.

The central research question addressed in this report is: To what extent do demand, wind, solar, interconnector flows, and system constraints explain and predict short-term price spreads between consecutive I-SEM markets, and how can this predictive power be converted into a robust arbitrage trading strategy?

3. Data Description and Preparation

3.1 Dataset Overview

The dataset, provided by Energia, contains half-hourly observations of market prices, system demand, wind and solar generation forecasts, interconnector flows, and related system metrics spanning January 2022 to December 2025. Following initial loading, the dataset comprised 72,945 rows and 105 columns.

3.2 Data Quality and Cleaning Steps

Duplicate observations lead to inflated model performance metrics. Hence, 2721 of them were identified and removed. A delivery timestamp was created with parsed datetime columns. Following the resolution of 144 duplicate timestamps (which arose from a split in the 'PredictedICFlow' column and were collapsed to single records using a NaN-safe mean), and the removal of rows missing essential market variables, the final analytical dataset comprised 70,006 observations.

A small number of entries for actual wind generation containing physically implausible negative values, representing fewer than 0.01% of observations, were clipped to zero. Missingness was found to follow a market-structural pattern: IDA2 and IDA3 prices are only available within the specific delivery windows during which those auctions operate (11:00–23:00 and 17:00–23:00 respectively). Solar forecasts for IDA2 and IDA3 contained missing values within their auction windows (overnight hours) and non-zero entries outside them, both being resolved. Wind forecast columns with incomplete coverage were filled using available alternative forecast sources from Meteo and EmSys providers in cascade order. Demand forecast columns were found to be perfectly clean.

An outlier diagnostic using a relaxed $3 \times \text{IQR}$ threshold revealed that extreme price observations constitute only about 2% of the dataset, and are economically significant market events, not errors, which were retained. Prices for the DAM ranged from –€30 to €705/MWh; for the Balancing Mechanism, the range was –€456 to €874/MWh, reflecting the high volatility characteristic of real-time electricity markets.

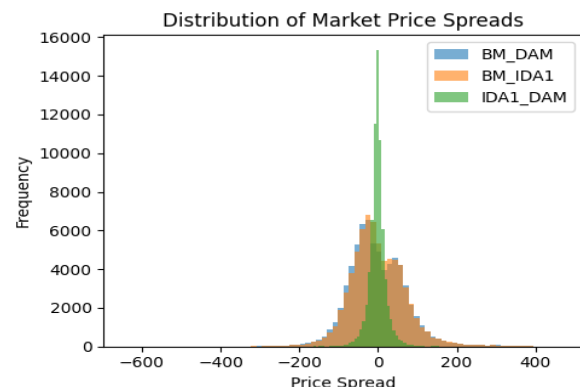
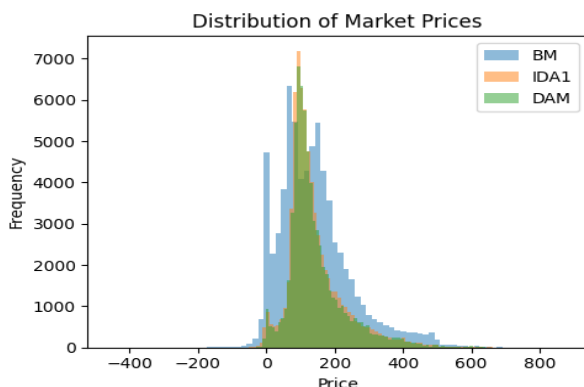
Market	Mean (€/MWh)	Std Dev (€/MWh)	Median (€/MWh)	Minimum (€/MWh)	Maximum (€/MWh)	Observations
DAM	142.86	84.57	119.00	–30.00	705.47	70,006
IDA1	141.23	84.91	117.77	–70.54	724.00	70,006
IDA2	152.77	89.33	127.26	–37.19	670.00	35,024
IDA3	166.33	92.39	142.00	–17.00	661.04	17,512
BM	141.32	109.07	126.11	–456.34	874.25	70,006

Table 1: Summary statistics for I-SEM market prices across the cleaned dataset.

4. Exploratory Data Analysis

4.1 Defining and Characterising Price Spreads

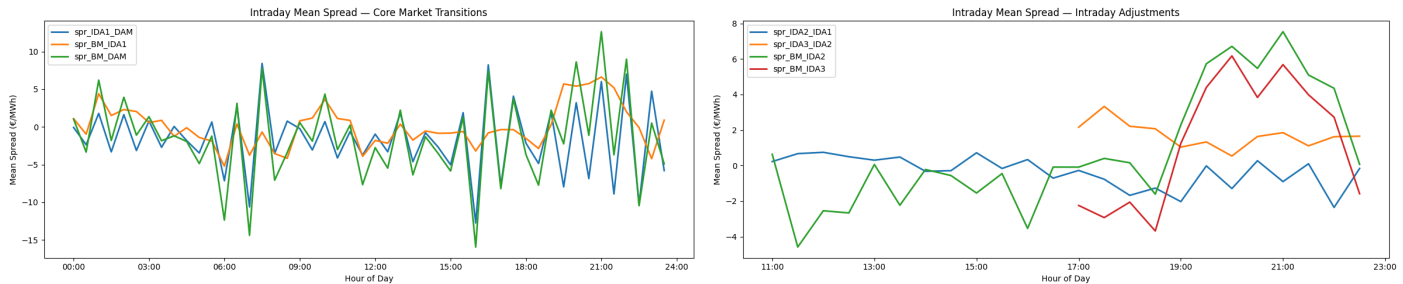
Seven price spreads were defined to capture the price difference: **DAM→IDA1**, **IDA1→IDA2**, **IDA2→IDA3**, **IDA1→BM**, **IDA2→BM**, **IDA3→BM**, and **DAM→BM**. An analysis of spread direction revealed that most spreads are roughly balanced between positive and negative outcomes: the DAM→IDA1, IDA1→BM, and DAM→BM spreads show near-50/50 splits. This is an important finding as profitable trading requires genuine predictive ability, not simply exploiting some market bias.



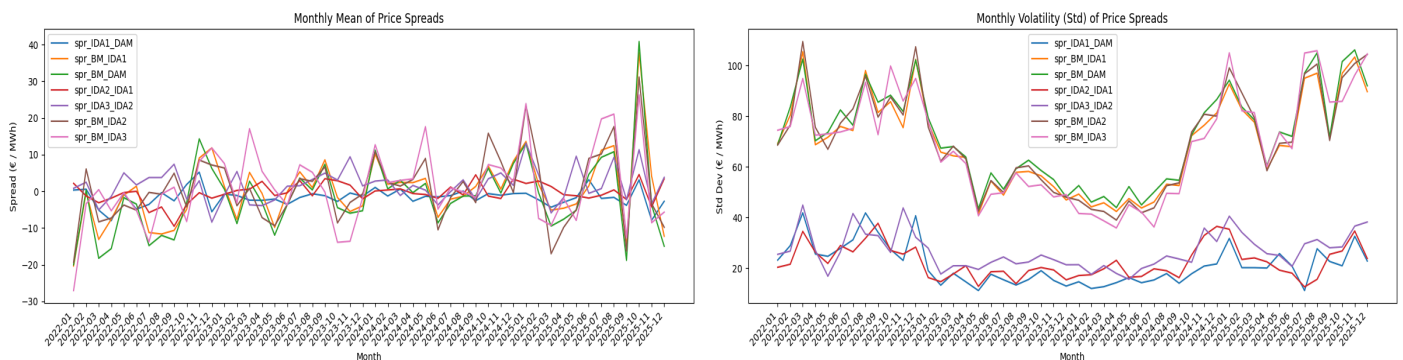
The spread histograms' extended tails suggest that some delivery intervals exhibit noticeably larger variances than the majority, which only exhibit minor differences. This suggests that potential gains may be concentrated in specific time periods rather than distributed evenly over time. However, the IDA1→IDA2 and IDA2→IDA3 spreads show a clear negative skew: later intraday auctions usually settle lower than earlier ones, around 76% and 87% of the time, respectively.

4.2 Intraday Patterns and Seasonality

Mean spread magnitudes and standard deviations are systematically higher during morning and evening peak hours. Spreads narrow during overnight hours. This time-of-day structure suggests that predictive models should use hour-of-day as an explanatory variable, since trading signals are likely to be more profitable during peak periods.



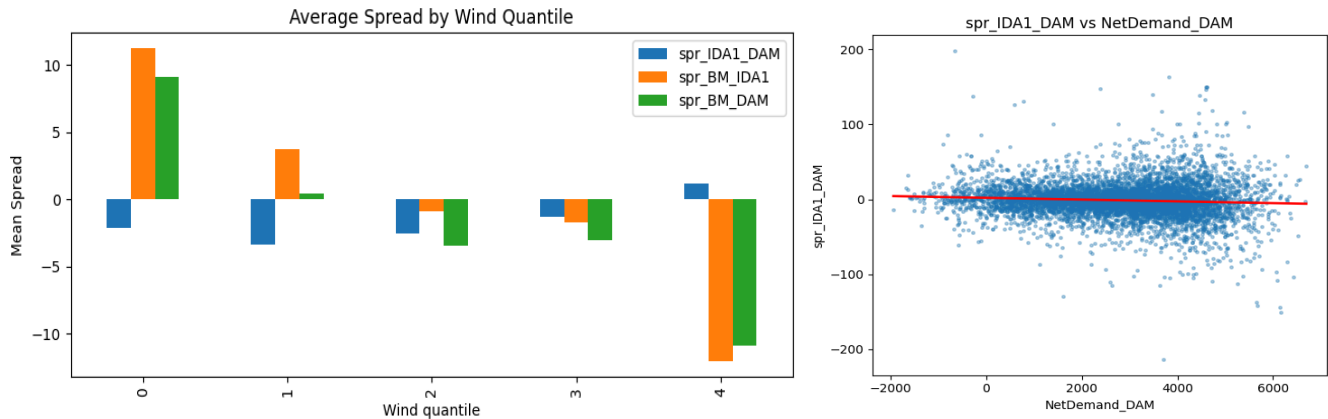
Reviewing monthly aggregates revealed that spread behaviour is not constant across the year. Volatility is high in certain months. This seasonal variation means that model performance may vary across time periods, and that out-of-sample testing should be treated cautiously, as previously observed volatility patterns may not repeat.



4.3 Relationship Between Spreads and System Fundamentals

Conditional analysis confirmed that price spreads are systematically related to underlying system conditions. When actual wind generation is low, the IDA1→BM spread averages approximately €11/MWh, when wind is high, the same spread averages approximately -€12/MWh. This confirms that renewable energy shifts the balance between markets and creates predictable and directional arbitrage opportunities.

Net demand, defined as total electricity demand minus wind and solar generation, shows a similarly structured relationship. Higher net demand is associated with larger spreads and higher market prices, while abundant renewable output suppresses spreads. Scatterplots and Pearson correlation map confirmed that no single variable determines spread outcomes, and that the relationships are not strictly linear. This non-linearity motivates the use of ensemble machine learning methods, which can capture intricate, higher-order interactions that linear models would overlook.



5. Predictive Modelling Methodology

5.1 Feature Engineering

A rich feature set was constructed to capture the information available to a trader at the gate closure of each market. The principle of no future information being leaked into model inputs was strictly enforced. Only variables known prior to each market's gate closure were included as features for that transition. Beyond the raw market and forecast variables, several derived features were engineered. Cyclical calendar values using sine and cosine transformations were used to represent **hour of day**, **day of week**, and **month of year** in a way that preserves their periodic nature. **Net demand** was computed for each auction horizon as demand minus wind and solar forecasts. **Wind forecast consensus** and **uncertainty** were computed as the mean and standard deviation across the multiple available forecast providers (Meteo, EmSys, EirGrid), with the spread between providers serving as a proxy for forecast confidence. **Forecast revision** variables capture the change in wind or demand estimates between successive auction horizons, which directly measures how new information has altered the system picture between one gate and the next. **Lag variables**₍₁₎ for spreads and prices over 1-day, 2-day, 7-day, and 14-day windows captured momentum and mean-reversion patterns.

5.2 Model Architecture and Validation

Two separate model types were trained for each market transition. A gradient-boosted classification model (**LightGBM**)₍₂₎ was used to predict the direction of the spread, whether the later market would settle above or below the earlier one. A gradient-boosted regression model was used to predict the magnitude of the spread. Both models were embedded in a preprocessing pipeline that imputed missing values using column medians and applied standard scaling. To respect the temporal ordering of the data and avoid look-ahead bias, all model evaluation was conducted using a strict train–test split: the first 80% of observations were used for training, and the final 20% were held back for evaluation. This is a critical methodological choice in time-series financial modelling, where random cross-validation would produce inflated performance estimates by allowing future data to inform historical predictions. A **five-fold walk-forward cross-validation** was additionally used to assess model stability across different time periods.

6. Model Performance Results

6.1 Classification Performance

Table 2 summarises the classification performance for all seven market transitions. The **DAM→IDA1** transition stands out as the most predictable, achieving 70.18% directional accuracy on the held-out test set

and a ROC-AUC of 0.7718, indicating meaningful discriminatory power well above the random baseline of 0.50. The five-fold walk-forward cross-validation mean of 69.76% ($\pm 2.0\%$) confirms that this performance is stable over time rather than concentrated in a single favourable period. The BM-related transitions: IDA1→BM, IDA2→BM, IDA3→BM, and DAM→BM are considerably harder to predict, with accuracies ranging from 55% to 59%. This is consistent with the nature of the Balancing Market, which reflects real-time system imbalances and is more responsive to events that are hard to forecast from the earlier available data.

Transition	Test Accuracy	ROC-AUC	CV Accuracy ($\pm$ Std)
DAM → IDA1	70.18%	0.7718	69.76% ($\pm 2.00\%$)
DAM → BM	58.54%	0.6161	56.99% ($\pm 2.20\%$)
IDA1 → BM	56.24%	0.5779	55.17% ($\pm 1.92\%$)
IDA1 → IDA2	57.18%	0.6178	56.72% ($\pm 1.98\%$)
IDA2 → BM	56.12%	0.5653	55.08% ($\pm 1.89\%$)
IDA2 → IDA3	55.74%	0.5862	55.12% ($\pm 2.01\%$)
IDA3 → BM	55.60%	0.5582	55.17% ($\pm 2.05\%$)

Table 2: Model performance summary across all seven market transitions; CV = cross-validation

7. Trading Strategy Design and Evaluation

7.1 Strategy Design

The direction of each trade is determined by the classification model: if the model assigns a probability above 0.5 that the spread will be positive; if below 0.5, the reverse. Only trades where the model indicates a strong enough opinion are retained after a confidence threshold parameter eliminates trades where the predicted probability is near 0.5. The regression model's predicted spread magnitude determines the size of the position, a larger spread is predicted by the model, which commits more volume to the trade and smaller forecast spreads draw smaller positions. Instead of treating all signals equally, this magnitude-proportional sizing makes sure that the strategy puts more money into the intervals that it deems most promising.

7.2 Performance Summary

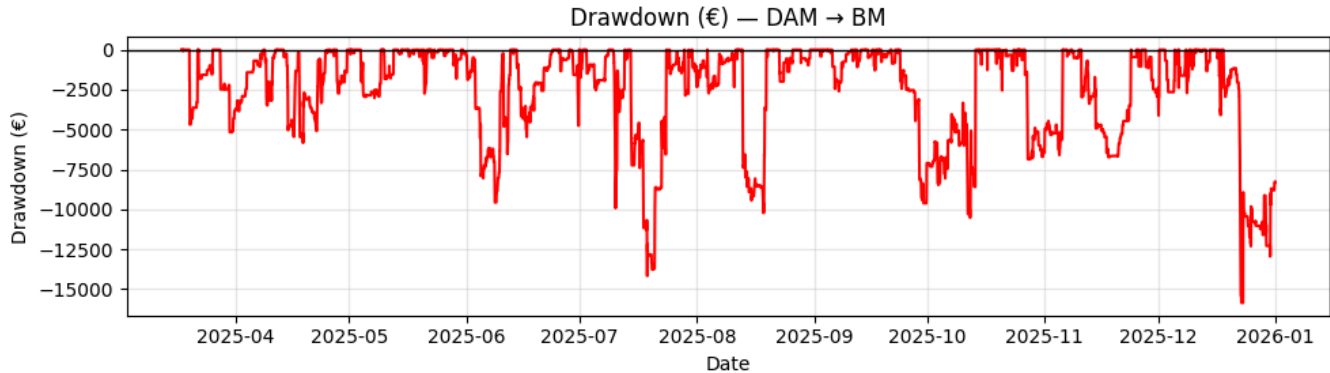
Table 3 presents the trading strategy results at the optimal confidence threshold of 0.20 (i.e., only trades where the model's probability exceeds 0.70 or falls below 0.30 are executed). This threshold was tuned by evaluating profitability metrics across few threshold levels and choosing the level that maximised earnings per MWh while preserving a viable trade count.

Transition	Test Size	Trades	Hit Rate (%)	€/MWh	Total P&L (€)
DAM → BM	13,783	4,258	70.36	33.75	480,887.04
IDA1 → BM	13,829	4,064	65.28	17.67	271,178.15
IDA2 → BM	6,928	1,850	63.84	17.36	95,731.73
DAM → IDA1	13,653	7,350	80.93	14.11	375,056.06
IDA3 → BM	3,467	907	59.21	9.64	33,689.05
IDA1 → IDA2	6,857	2776	58.43	7.18	83,058.07
IDA2 → IDA3	3,417	952	59.98	7.96	26,469.34

Table 3: Trading strategy results at confidence threshold 0.20

7.3 Interpreting the Results

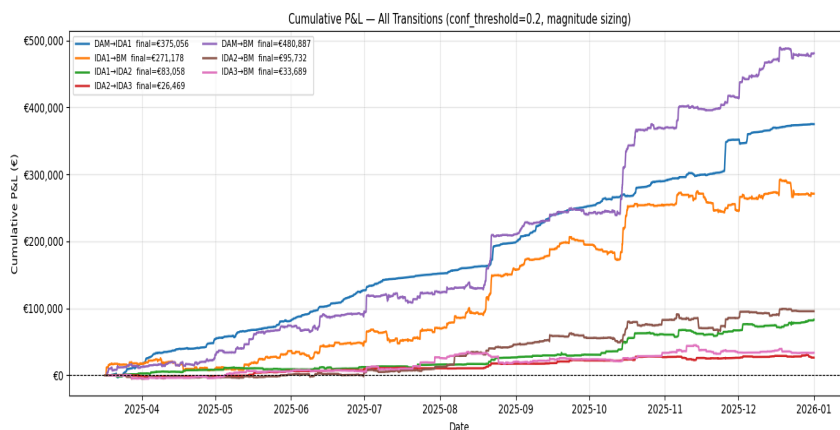
The **DAM→IDA1** transition offers the clearest and most reliable arbitrage opportunity. Although its total earnings (€375,056.06) and earnings per MWh (€14.11) are lower than the **DAM→BM** transition, it achieves the highest hit rate of **80.93%** at the 0.20 confidence threshold, with the greatest number of executed trades (**7,350**). The **DAM→BM** transition, while the most profitable on a per-MWh basis, carries significantly higher risk. Its earnings per MWh is €33.75 and total profit is €480,887.04. The maximum drawdown for DAM→BM



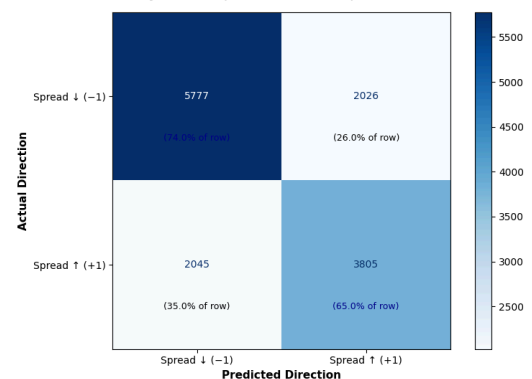
over the test period was approximately €15,852.23.

8. Conclusions and Recommendations

This analysis demonstrates that I-SEM price spreads between consecutive markets are not random, but are systematically influenced by observable system fundamentals. Based on the findings, the recommended trading strategy for Energinet would centre on the **DAM→IDA1** transition as the primary instrument, given its superior accuracy and high trade count. The **DAM→BM** transition could be run in parallel at reduced volume as a supplementary strategy, subject to a more detailed risk evaluation. Model performance should be monitored on a rolling basis, with accuracy and profitability assessed monthly. Given the observed seasonality in spread behaviour, consideration should be given to season-specific model variants or the inclusion of explicit seasonal interaction terms.



Confusion Matrix — DAM→IDA1
Accuracy: 70.18% | Recall ↓: 74.0% | Recall ↑: 65.0%



References and Data Sources

Energinet (2025). MarketData_2022-2026.parquet. Proprietary dataset provided for Analytathon assignment.

(1) Uniejewski, Bartosz & Marcjasz, Grzegorz & Weron, Rafał. (2019). Understanding intraday electricity markets: Variable selection and very short-term price forecasting using LASSO. International Journal of Forecasting. 35. 10.1016/j.ijforecast.2019.02.001.

(2) Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and, Liu, T.-Y. (2017) LightGBM: A Highly Efficient Gradient Boosting Decision Tree. Proceedings of the 31st International Conference on Neural Information Processing Systems, Long Beach, CA, December 2017, 3149-3157.